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Daily Thermal Energy Storage in Water-in-Glass Evacuated-Tube Solar Water Heaters: A Paired Experimental Assessment of an Aluminum-Foil Reflector

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Abstract: Improving the thermal performance of solar water heating systems is an important aspect of sustainable energy use, particularly in applications where solar heat can reduce reliance on conventional water heating. Reflective surfaces can increase the solar radiation reaching evacuated-tube collectors, but their contribution to daily thermal energy storage requires evaluation under comparable operating conditions. This study investigates the difference in daily stored thermal energy between water-in-glass evacuated-tube solar water heaters operated with and without an aluminum-foil reflector assembly. Three systems were tested simultaneously in Trujillo, Peru, including two nominally identical 316 L prototypes with and without a 6.624 m^2 aluminum-foil-covered steel reflector and a commercial heater (C) used as a secondary reference. Tank temperatures were measured at three heights, and daily stored-energy gains were calculated over a common monitoring interval of 06:25–18:50 without water withdrawal or replenishment. Twelve paired daily observations were analyzed to assess the differences between the two prototypes. The reflector-equipped prototype (PR) recorded greater stored-energy gains in all 12 observations, with mean daily values of 34.39 MJ compared with 30.56 MJ for the prototype without a reflector (P0). The mean paired difference was 3.83 MJ, corresponding to an overall increase of 12.54%, while individual daily increases ranged from 9.18% to 15.61%. The 95% confidence interval for the mean paired difference was 2.81–4.85 MJ. C recorded a mean daily gain of 26.29 MJ, although differences in capacity and collector geometry prevented a direct assessment of the reflector effect from this comparison. The findings indicate that the aluminum-foil reflector assembly is associated with a consistent increase in daily stored thermal energy under the tested conditions. The study provides experimental evidence for evaluating a relatively simple modification to evacuated-tube solar water heating systems, while distinguishing improvements in stored thermal energy from changes in collector efficiency.

Keywords: Solar water heating; Water-in-glass evacuated-tube collector; Aluminum-foil reflector; Thermal energy storage; Experimental performance assessment; Renewable energy utilization

1 Introduction

Solar water heating provides a practical means of supplying low-temperature heat from renewable energy, particularly in residential and commercial buildings where hot-water demand accounts for a substantial share of thermal energy consumption. The contribution of these systems to sustainable heat supply depends on both the collection of solar radiation and the retention of useful heat for periods when solar input is reduced. Evacuated-tube collectors are widely used for this purpose because vacuum insulation limits heat losses from the absorbing surface. Recent studies have examined advances in evacuated-tube design, thermal performance, energy storage, and system integration, emphasizing the relationship between collector configuration and practical energy utilization [1, 2]. Improving the amount of thermal energy accumulated in the storage tank is therefore an important aspect of solar water heater development.

Water-in-glass evacuated-tube solar water heaters combine solar collection and sensible heat storage through natural circulation between the tubes and the tank. Solar heating creates density differences within the water, establishing thermosiphon circulation without mechanical pumping. The thermal response of the system depends on incident radiation, tube spacing, collector inclination, circulation characteristics, and temperature distribution within the storage tank. Morrison et al. [3] investigated water circulation in this collector configuration through experimental measurements and numerical simulation, establishing the relationship between flow behavior and the thermal state of the collector–storage system. More recently, Mendoza Orbegoso et al. [4] examined the influence of heat-flux symmetry and inclination on the thermofluidic behavior of water-in-glass evacuated tubes using computational fluid dynamics. These studies provide a physical basis for assessing how changes in solar radiation received by the tubes influence heat transfer and energy accumulation in the connected storage tank.

Research on evacuated-tube solar collectors has followed several approaches to improving thermal performance. Gong et al. [5] combined numerical simulation and experimental analysis to investigate an all-glass heat-pipe evacuated collector tube, including the use of a reflective film to modify its optical performance. Yang et al. [6] examined the optical and thermal performance of a compact collector incorporating heat-pipe evacuated tubes. Zhang et al. [7] experimentally compared two evacuated-tube collectors using flat heat pipes and different internal heat transfer arrangements. These investigations show that collector performance can be improved through modifications to the absorber, internal heat transfer components, and optical configuration. They also underline the importance of identifying the performance variable being evaluated, since collector efficiency, outlet temperature, and stored thermal energy describe different aspects of system operation.

Recent experimental work has extended these investigations to complete solar water heating systems. Khadom et al. [8] evaluated evacuated-tube solar water heaters incorporating preheating units, considering thermal performance and environmental implications. Uniyal et al. [9] experimentally investigated a U-tube evacuated-tube collector incorporating phase change material and wire mesh for hot-water generation. Elbrashy et al. [10] examined the thermal behavior of a storage material integrated into an evacuated-tube collector through experimental measurements and mathematical analysis. Zhu et al. [11] experimentally investigated an air-type solar collector incorporating phase change material within evacuated tubes. Mellouli et al. [12] developed a three-dimensional numerical model to examine a photovoltaic–thermal collector combining evacuated tubes and phase change materials, including the effects of reflector arrangements and operating conditions. Together, these investigations demonstrate that collector design and thermal storage arrangements must be evaluated in relation to the intended energy output and operating conditions.

Reflective surfaces provide another route to modifying the energy received by a solar collector. A reflector positioned beneath or adjacent to an absorber can redirect part of the incident radiation toward the receiving surface. Its effect depends on the relative geometry of the reflector and collector, the angular distribution of solar radiation, and the optical properties of the reflective material. Mandal and Ghosh [13] experimentally investigated a double-pass solar water heater equipped with a reflector. Nikolić et al. [14] examined the dimensions and positioning of a planar back reflector associated with a bifacial solar collector, demonstrating the importance of the geometrical relationship between the reflective surface and the receiving area. Although these collector configurations differ from water-in-glass evacuated-tube systems, they establish relevant principles for evaluating external reflector arrangements.

The application of reflectors to evacuated-tube systems has also received attention in recent research. Farghaly et al. [15] investigated evacuated-tube collectors coupled with parabolic reflectors in a solar desalination system. Sheikholeslami and Al-Jammasi [16] studied an integrated U-tube collector incorporating a parabolic reflector and phase change thermal storage. Zhu et al. [17] examined a concentrating solar thermal arrangement combining an evacuated-tube absorber with heat storage rods. These studies examine different approaches to improving solar thermal performance, including reflector-assisted configurations [15, 16] and heat storage elements integrated with evacuated-tube absorbers [17]. The collector geometries, heat transfer arrangements and performance indicators examined in these studies differ from those considered here. Results expressed in terms of thermal efficiency, freshwater production, or phase change storage performance cannot be directly equated with the increase in sensible energy stored in a water-in-glass solar water heater.

The optical behavior of the reflective material is equally important. Aluminum foil provides a readily available surface for constructing a reflector, although its reflection characteristics depend on surface finish and the distribution of incident radiation. Pozzobon et al. [18] measured the reflectivity of the bright and matte sides of household aluminum foil and identified differences in their specular and diffuse reflection characteristics. For an evacuated-tube array, the radiation redirected toward the absorber also depends on tube spacing, reflector position, and solar incidence angle. Consequently, the thermal response of an installed reflector assembly is determined by the interaction between its material properties and the collector geometry. A system-level experimental comparison provides a direct means of evaluating the resulting change in stored thermal energy.

The relationship between collector performance and thermal storage is particularly relevant to domestic solar water heating. The useful energy retained in a storage tank depends on the water mass and the temperature change

over the assessment period. Temperature stratification introduces a further consideration because the thermal state of the tank may differ substantially between measurement heights. Studies of integrated evacuated-tube storage systems [9–12] have examined these interactions using different heat transfer and storage arrangements. For direct water-in-glass circulation, an energy-based comparison is particularly appropriate because the storage tank forms part of the operating collector system. The highest water temperature reached during a test does not necessarily correspond to the greatest increase in stored energy when the compared systems contain different water volumes.

Collector geometry also affects the interpretation of experimental performance comparisons. Differences in tube spacing, inclination, and storage capacity can influence the measured thermal response independently of an added reflector. Mendoza Orbegoso et al. [19] developed and experimentally validated a predictive framework for water-in-glass evacuated-tube solar heaters, incorporating shading, collector geometry, and thermosiphon behavior. Their work demonstrated the importance of these design parameters in determining system performance. The instrumentation and real-time monitoring framework used for experimental assessment was documented by Alva Alcántara et al. [20]. These investigations provide the modeling and measurement background for examining the thermal response of geometrically comparable prototypes.

The distinction between collector-level improvement and system-level energy storage is also relevant to sustainable solar thermal applications. Singh [21] examined energy and environmental performance in a solar desalination-assisted water heating arrangement, while Agrawal and Kumar [22] evaluated an evacuated-tube-based atmospheric water harvesting system with and without phase change storage. Hemmatian et al. [23] investigated evacuated-tube-assisted solar still configurations incorporating thermal storage, and Hashim et al. [24] examined the operating performance of an evacuated-tube receiver under different flow conditions. These studies illustrate the range of performance measures used in solar thermal applications, including thermal output, energy efficiency, and environmental indicators. For the present study, the relevant measure is the increase in sensible thermal energy stored in the water during a defined daily interval.

Recent studies have also examined the combined influence of operating conditions and collector design on thermal performance. Farhadi et al. [25] developed analytical and data-driven models for heat-pipe evacuated-tube collectors, considering geometrical, optical, and thermohydraulic parameters. Ozsoy and Galip [26] experimentally evaluated passive elements in evacuated U-pipe collectors and examined their effects on thermal performance. Although these systems employ different circulation arrangements from water-in-glass heaters, they reinforce the importance of comparing collector modifications using clearly defined operating conditions and performance measures.

A further consideration is the separation of individual design modifications from the performance of the complete system. When a reflector is introduced together with changes in tube arrangement, circulation, or storage components, the measured energy difference represents the combined configuration. A more focused comparison can be obtained by operating two prototypes with the same nominal geometry and water inventory simultaneously, with the reflector assembly as the principal intentional construction difference. Repeated daily observations then provide a basis for examining the magnitude and consistency of the stored-energy difference under changing outdoor conditions. This approach also distinguishes the primary reflector comparison from a broader assessment involving commercial heaters with different design characteristics.

The present study addresses this experimental comparison using water-in-glass evacuated-tube solar water heaters operated simultaneously in Trujillo, Peru. Three systems were examined: a prototype without a reflector (P0), a reflector-equipped prototype (PR), which is equipped with an aluminum-foil-covered steel reflector, and a commercial heater (C). Both experimental prototypes shared the same nominal tube arrangement, inclination, and storage-tank construction, with a water filling volume of 316 L each. The reflector consisted of a steel backing plate covered with aluminum foil, providing an exposed surface area of 6.624 m^2 beneath the evacuated-tube array. C was retained as a secondary reference because its storage capacity and collector geometry differed from those of the prototypes.

The analysis focuses on the daily increase in sensible thermal energy stored in the tank water over a common monitoring interval of 06:25–18:50. Twelve matched daily observations were used to quantify the absolute and relative stored-energy differences between PR and P0. Temperature measurements at three tank heights were combined with daily energy calculations and paired statistical comparisons to assess the thermal response of the reflector-equipped configuration. The simultaneous experimental arrangement maintained common outdoor conditions for the primary comparison and allowed the commercial system to be evaluated separately.

The main contribution of this study is the experimental quantification of the daily stored-energy difference associated with adding an aluminum-foil-covered reflector to a water-in-glass evacuated-tube solar water heater while maintaining the nominal collector geometry and water inventory. Unlike studies that combine several structural or thermal modifications, the paired comparison focuses on the installed reflector assembly. By evaluating stored sensible energy rather than collector temperature or efficiency alone, the study connects reflector-assisted collector design with the thermal energy retained for domestic water heating. The findings provide a quantitative basis for assessing practical reflector modifications intended to improve renewable heat utilization in solar water heating systems.

2 Experimental Setup and Methodology

2.1 Study Site and Experimental Design

The experimental campaign was conducted in Trujillo, Peru (approximately 8.11° S, 79.03° W), at an elevation of 34 m above sea level, from 31 December 2024 to 26 February 2025. The present analysis covers 12 matched daily thermal-energy observations collected during the comparison period from 25 January to 26 February 2025. All measurements were obtained under outdoor operating conditions, allowing the thermal response of the tested systems to be examined under the same daily weather conditions.

Three water-in-glass evacuated-tube solar water heaters were installed at the experimental site: a P0, a prototype incorporating a PR, and a C. The three systems were operated simultaneously throughout each daily test. This arrangement provided a common basis for comparing their temperature development and stored thermal energy under concurrent solar and ambient conditions.

The principal experimental comparison was between P0 and PR. Both prototypes were manufactured using the same materials, collector dimensions, tube arrangement, inclination and storage-tank construction. Each was filled with 316 L of water, and equal water inventories were maintained during the tests. The addition of the aluminum-foil-covered steel plate to PR was the only intentional construction difference between the prototypes. The paired design therefore allowed the thermal response associated with the installed reflector assembly to be assessed while keeping the principal geometrical and operating parameters consistent.

C was included as a secondary reference for system-level performance. Its collector geometry and storage capacity differed from those of the experimental prototypes, and its results were therefore evaluated separately from the primary reflector comparison. One physical unit represented each configuration. The 12 matched observations corresponded to repeated daily tests of these units rather than independent replication across separately manufactured heaters.

2.2 Heater Construction and Geometry

Each solar water heater was equipped with 30 double-glass evacuated tubes, each 1.80 m long. The outer diameters of the inner and outer concentric glass tubes were 48 and 58 mm, respectively. The two experimental prototypes had a center-to-center tube spacing of 116 mm and a collector inclination of 15° relative to the horizontal. C had a smaller tube spacing of 84 mm and a greater inclination of 25°, with a reported storage capacity of 263 L. Both experimental prototypes were operated with a water filling volume of 316 L.

The storage tanks consisted of a 1 mm thick AISI 304 stainless-steel inner wall, a 50 mm polyurethane insulation layer and a 0.4 mm stainless-steel outer cover. The centerline of each prototype storage tank was positioned approximately 1.20 m above the supporting base. Water circulated naturally between the evacuated tubes and the storage tank through thermosiphon action, driven by density differences resulting from solar heating. The arrangement required no mechanical circulation pump.

Figure 1 shows C and PR, while Figure 2 presents the prototype geometry and principal structural details. The common collector geometry and storage-tank construction of P0 and PR established the basis for comparing the daily stored-energy response of the two configurations.

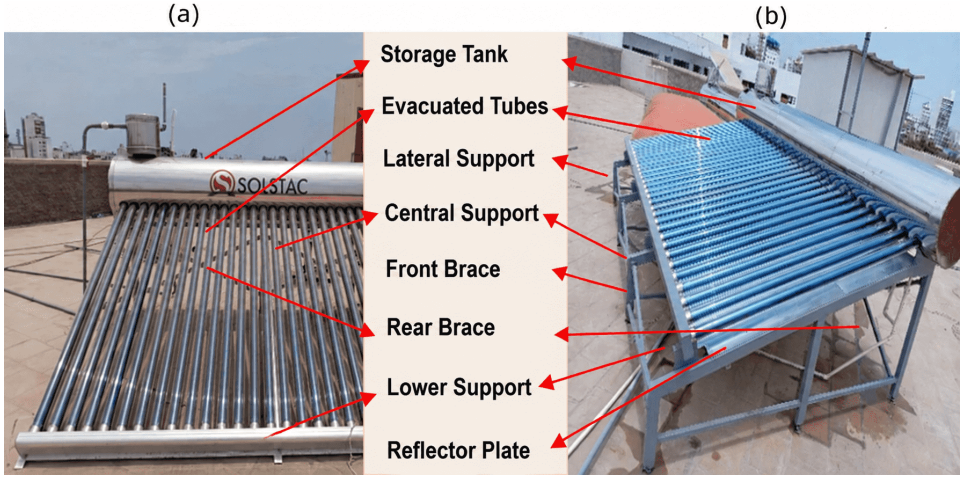


Figure 1. Experimental solar water heater configurations: (a) C and (b) PR, showing the main structural components

Note: C denotes the commercial heater and PR denotes the reflector-equipped prototype.

2.3 Reflector Assembly

The reflector assembly comprised a rectangular steel backing plate covered with aluminum foil attached using an adhesive. The exposed reflective surface measured 3.680 m × 1.800 m, giving a total area of 6.624 m^2. The plate was positioned beneath the evacuated-tube array at a reported distance of 90 mm from the tube centerline, as illustrated in Figure 2.

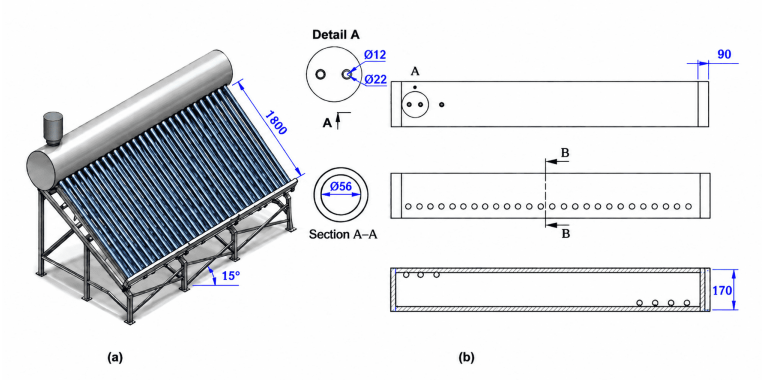


Figure 2. Experimental prototype geometry and structural details: (a) overall arrangement of the evacuated-tube solar water heater and (b) storage-tank views and sectional details

The aluminum foil was installed with its brighter surface facing the evacuated tubes. The steel plate provided structural support, while the exposed aluminum surface served to redirect incident solar radiation toward the tube array. The reflector was incorporated without changing the nominal tube arrangement, collector inclination or storage volume of the prototype.

The installed assembly was evaluated as a complete modification to the collector configuration. Its performance was determined from the measured thermal response of the water stored in the tank rather than from separately assigned optical properties of the foil. The solar-weighted reflectance, thermal emittance and angular reflection characteristics of the installed surface were not measured. Accordingly, the analysis was based on the observed stored-energy difference between PR and P0, encompassing the combined thermal effects associated with the plate-and-foil assembly.

2.4 Instrumentation and Monitoring

The thermal response of each heater was monitored through water-temperature measurements at three vertical positions in the storage tank. Three-wire Pt100 resistance temperature detectors were installed in the lower, middle and upper regions using a sensor support inserted into the tank, as shown in Figure 3. This arrangement provided simultaneous temperature measurements at different heights and allowed the development of thermal stratification to be examined during the daily heating period. The instrumentation and monitoring architecture were documented in the related experimental monitoring study [20].

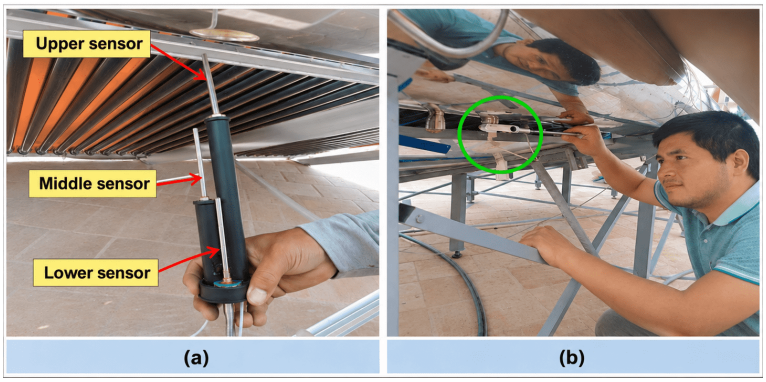


Figure 3. Pt100 temperature-sensor arrangement and installation: (a) upper, middle, and lower sensor positions; (b) installation of the sensor support at the bottom of the storage tank

At each recording time, the representative tank-water temperature was calculated as the arithmetic mean of the

three measured temperatures:

$$T_{\text{mean}}(t) = \frac{T_{\text{lower}}(t) + T_{\text{middle}}(t) + T_{\text{upper}}(t)}{3} \tag{1}$$

where, $T_{\text{lower}}(t)$, $T_{\text{middle}}(t)$, and $T_{\text{upper}}(t)$ are the water temperatures measured at the lower, middle and upper tank positions at time (t), respectively. The same averaging procedure was applied to all three heaters. This three-point arithmetic mean was used as the representative tank-water temperature in the stored-energy calculations. Because the measurements were not weighted by the water volume represented by each sensor, the calculated temperature remained an approximation of the volume-averaged tank temperature, particularly under stratified conditions.

The Pt100 sensors had a specified operating range of 0–100 ℃ and a reported precision of 0.1 ℃, according to the instrumentation specifications provided in [20]. Temperature signals were conditioned using Siemens SITRANS TH200 transmitters with 4–20 mA outputs and recorded through a Siemens S7-1200 programmable logic controller (PLC). A Siemens KTP700 Basic human–machine interface provided local visualization of the monitored variables. The reported sensor precision was treated as an equipment specification rather than a calibration-derived uncertainty for the complete measurement system.

Meteorological conditions were monitored during the experimental campaign. Global solar irradiance was measured using a photovoltaic pyranometer with a reported measurement range of 0–1800 W/m^2, a resolution of 1 W/m^2 and a 0–10 V output. Wind speed was recorded using an anemometer with a measurement range of 0–30 m/s, a resolution of 0.1 m/s and a 0–10 V output. Ambient temperature was also monitored. The meteorological instruments and data-acquisition equipment are presented in Figure 4.

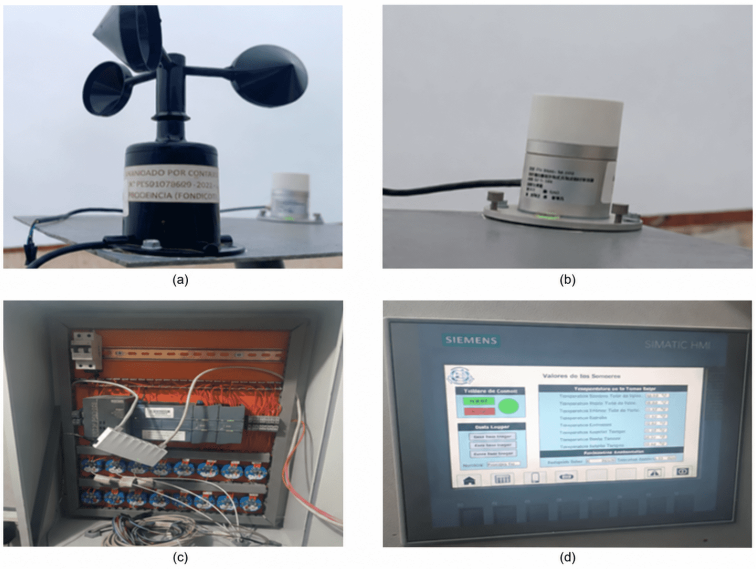


Figure 4. Meteorological monitoring and data-acquisition equipment: (a) anemometer, (b) pyranometer, (c) PLC-based control and data-acquisition cabinet, and (d) HMI

Note: PLC denotes the programmable logic controller and HMI denotes the human–machine interface.

2.5 Daily Experimental Procedure

Each daily experiment was conducted from 06:25 to 18:50 local time, corresponding to a monitoring interval of 12 h and 25 min. All three heaters operated simultaneously during this period. Water was neither withdrawn nor replenished during the daily measurements, so the recorded temperature changes represented the thermal response of the water inventory retained within each system.

At the end of each test, the final temperature was taken as the value recorded at 18:50 rather than the maximum temperature attained during the day. The initial temperature corresponded to the measurement at 06:25. This consistent assessment interval allowed daily changes in stored thermal energy to be calculated using the same starting and closing times for all three configurations.

After completion of the daily test, the water in both experimental prototypes was replaced during the night in preparation for the following experimental day. The filling procedure maintained the same water volume of 316 L in P0 and PR. Since the initial water temperatures could differ between the two systems, the energy assessment was based on the temperature change measured separately for each heater. This procedure preserved the paired comparison while accounting for the individual initial and final thermal states of the systems.

2.6 Stored Thermal-Energy Assessment

The principal performance indicator was the daily increase in sensible thermal energy stored in the tank water. The calculation used the water mass and the change in representative tank temperature between the beginning and end of each daily test. The temperature dependence of the specific heat capacity of water was expressed through the following relation:

$$Q_j = m \int_{T_{i,j}}^{T_{f,j}} c_p(T) \mathrm{d}T \tag{2}$$

where, Q_j denotes the daily change in sensible thermal energy stored in the tank water on experimental day (j), m is the water mass, $c_p(T)$ is the specific heat capacity of water at temperature T, and T_{ij} and T_{fj} are the representative tank-water temperatures recorded at 06:25 and 18:50, respectively. The two experimental prototypes were operated with the same water filling volume of 316 L. The stored-energy calculation was based on the water mass and the temperature change of each heater over the specified monitoring interval. The resulting energy values were expressed in MJ.

The representative temperatures were obtained using Eq. (1). The same calculation procedure was applied to all heater configurations, with the corresponding water inventory used for each system. For the two experimental prototypes, the equal filling volumes provided a consistent basis for comparing the calculated energy gains.

The calculated Q_j represents the net increase in sensible energy stored in the tank water over the daily assessment interval. It reflects the combined effects of solar heat input, thermal transfer within the system and heat losses during operation. The calculation does not represent the total solar energy incident on the collector or the collector's thermal efficiency. The use of three-point mean temperatures also means that the stored-energy values are estimates based on the adopted temperature-sampling method.

2.7 Performance Comparison and Statistical Analysis

The performance assessment was based on 12 matched daily observations for each heater configuration. The principal comparison involved PR and P0, which were operated simultaneously with the same nominal collector geometry and water inventory. For each experimental day (j), the absolute difference in stored thermal energy was calculated as:

$$\Delta Q_j = Q_{\mathrm{PR},j} - Q_{\mathrm{P0},j} \tag{3}$$

where, $Q_{\mathrm{PR},j}$ and $Q_{\mathrm{P0},j}$ are the daily stored-energy increases of the reflector-equipped and unmodified prototypes, respectively.

The relative difference for each day was calculated using P0 as the reference:

$$G_j(\%) = \frac{Q_{\mathrm{PR},j} - Q_{\mathrm{P0},j}}{Q_{\mathrm{P0},j}} \times 100 \tag{4}$$

The overall relative difference was determined from the mean daily stored-energy values:

$$G_{\text{overall}}\,(\%) = \frac{\bar{Q}_{\mathrm{PR}} - \bar{Q}_{\mathrm{P0}}}{\bar{Q}_{\mathrm{P0}}} \times 100 \tag{5}$$

where, $\bar{Q}_{\mathrm{PR}}$ and $\bar{Q}_{\mathrm{P0}}$ are the arithmetic means of the 12 daily stored-energy observations for PR and P0, respectively. The overall relative difference was reported separately from the arithmetic mean of the individual daily percentages, since the two indicators apply different weighting to the observed energy gains.

Descriptive statistics included the arithmetic mean, sample standard deviation, minimum and maximum values, and cumulative sum of the daily stored-energy increases. The paired energy differences were used to assess the performance of PR relative to the unmodified prototype.

A two-sided paired Student's t-test was applied to the 12 daily differences to evaluate whether the mean difference departed from zero. The corresponding 95% confidence interval was calculated using the Student's t distribution. An exact two-sided sign test was also performed to assess the consistency of the direction of the paired differences without requiring a normal distribution. Statistical significance was evaluated at $\alpha = 0.05$.

The statistical unit was the experimental day, with one paired stored-energy observation obtained from PR and P0 for each retained test. Measurements collected within the same day were not treated as independent replicates. The inferential analysis was based on the assumption that the paired differences from different experimental days were independent, while the t-based procedure additionally assumed an approximately normal distribution of those differences. The resulting statistical estimates described variation among the retained daily observations rather than calibration uncertainty or variation among independently manufactured heaters.

Comparisons involving C were evaluated descriptively. Because C differed from the prototypes in storage capacity, tube spacing and inclination, these comparisons characterized the relative stored-energy performance of the complete heater configurations. The principal assessment of the reflector assembly remained the paired PR–P0 comparison.

3 Results and Discussion

3.1 Daily Thermal Response of the Three Heater Configurations

Figure 5 presents the temperature profiles recorded at the upper, middle and lower positions of the storage tanks on 25 February 2025. The three panels correspond to C, P0 and PR, respectively. Global solar irradiance is also shown to relate the temperature development of each heater to the prevailing solar conditions.

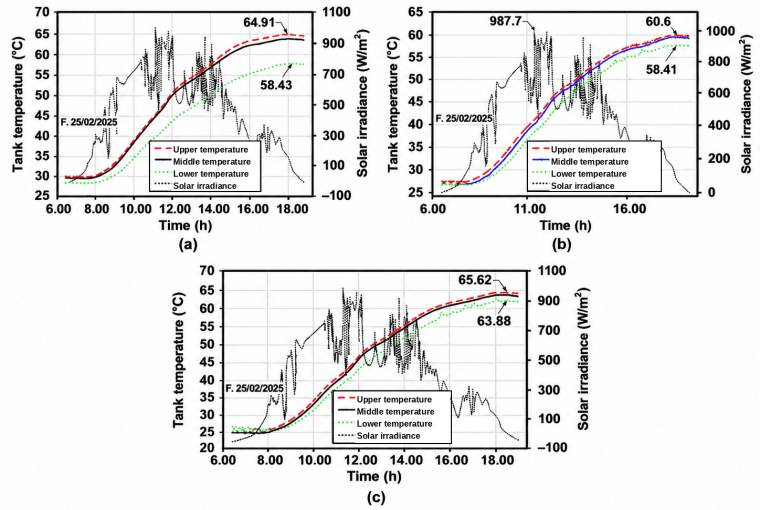


Figure 5. Temperature profiles at the upper, middle and lower positions of the storage tanks, together with global solar irradiance, on 25 February 2025: (a) C, (b) P0, and (c) PR

Note: C denotes the commercial heater, P0 denotes the prototype without a reflector, and PR denotes the reflector-equipped prototype.

All three systems exhibited a progressive increase in water temperature during the daytime heating period. The temperature curves responded more gradually than the fluctuations in solar irradiance, reflecting the thermal inertia of the stored water and the continuous exchange of heat between the evacuated tubes and the storage tanks. Toward the end of the monitoring interval, the rate of temperature increase declined as the systems approached their late-day thermal states.

C showed a greater separation between the upper and lower tank temperatures than either experimental prototype. In contrast, P0 and PR exhibited more closely grouped temperature profiles at the three measurement positions. This difference indicates a more uniform temperature distribution at the monitored heights in the experimental prototypes. The observed behavior is consistent with the interaction between natural circulation and thermal stratification in water-in-glass evacuated-tube systems, as discussed by Morrison et al. [3] and Mendoza Orbegoso et al. [4].

PR reached higher temperatures than P0 at the sampled tank heights during the later part of the heating period. Since the two prototypes shared the same nominal geometry and water inventory, this difference is consistent with the greater daily stored-energy gain associated with the reflector assembly. The temperature profiles also show why the performance of the three heaters should be evaluated using stored thermal energy rather than final temperature alone. The commercial unit contained 263 L of water, compared with 316 L in each prototype, so differences in water mass and temperature rise must both be considered.

3.2 Comparison of Mean Tank Temperature and Stored Thermal Energy

Figure 6 and Figure 7 compare the thermal responses of the three heaters during the experiment conducted on 25 February 2025. Figure 6 shows the arithmetic mean tank-water temperatures calculated from the three measurement positions, while Figure 7 presents the corresponding changes in stored thermal energy. The simultaneous measurements allow differences between the systems to be examined under the same daily solar and ambient conditions.

The mean-temperature profiles in Figure 6 show that PR maintained higher water temperatures than P0 during the later heating period. C reached a higher final mean temperature than P0, although this temperature ordering differed from the ranking obtained from stored thermal energy. This distinction arose because the heaters had different water inventories and temperature changes over the assessment interval.

The stored-energy profiles in Figure 7 followed the order PR > P0 > C. The reported daily gains were approximately 47.4 MJ for PR, 41.0 MJ for P0 and 36.2 MJ for C. Relative to C, P0 stored approximately 4.8 MJ more energy, corresponding to an increase of 13.3%, while PR stored approximately 11.2 MJ more, corresponding to an increase of 30.9%. The difference between the two experimental prototypes was approximately 6.4 MJ, equivalent to 15.6% of the energy stored by P0.

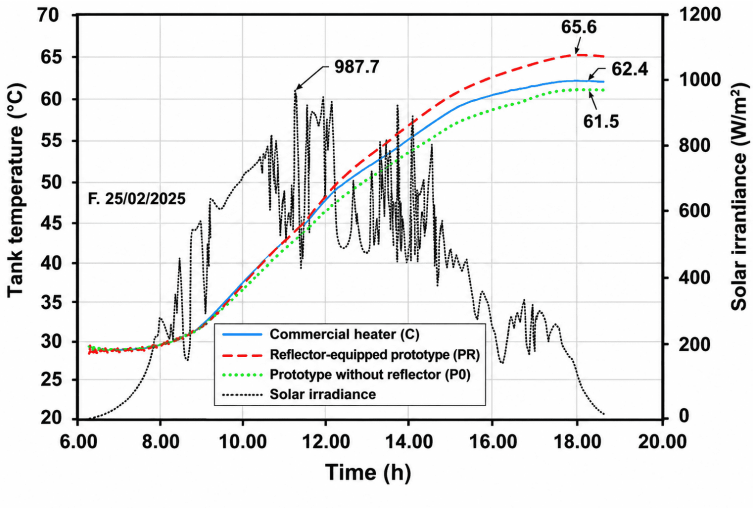


Figure 6. Mean tank-water temperatures of C, P0, and PR, together with global solar irradiance, on 25 February 2025

Note: C denotes the commercial heater, P0 denotes the prototype without a reflector, and PR denotes the reflector-equipped prototype.

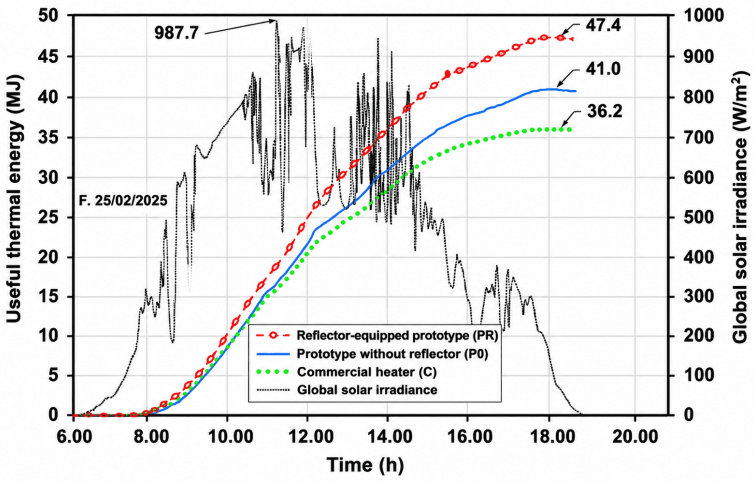


Figure 7. Development of stored thermal energy and global solar irradiance on 25 February 2025 for C, P0, and PR

Note: C denotes the commercial heater, P0 denotes the prototype without a reflector, and PR denotes the reflector-equipped prototype.

The comparison between PR and P0 is particularly relevant because both prototypes had the same nominal collector geometry and contained equal water inventories. Their difference therefore provides a more direct measure of the thermal response associated with the installed reflector assembly. In contrast, the comparison with C reflects differences in storage capacity, tube spacing and collector inclination in addition to the presence or absence of the reflector.

The results also distinguish the temperature attained by a heater from the thermal energy accumulated in its storage tank. A higher final water temperature does not necessarily correspond to a greater stored-energy increase when the systems contain different quantities of water. The energy-based assessment therefore provides a more appropriate basis for comparing the daily thermal performance of the tested configurations.

The profiles shown in Figures 5–7 represent one experimental day and provide a detailed illustration of the heating process. The corresponding daily energy values form part of the retained experimental observations and are not treated as additional independent measurements.

3.3 Daily Stored Thermal Energy Across the Paired Observations

The daily stored-energy gains obtained from the 12 matched experimental observations are presented in Table 1 and Figure 8. PR recorded the highest stored-energy increase in every observation, followed by P0 and C. This consistent ordering was maintained despite the substantial variation in daily energy accumulation across the experimental period.

Table 1. Daily increases in stored thermal energy for the three solar water heater configurations

Observation	PR (MJ)	P0 (MJ)	C (MJ)
1	37.77	34.05	28.69
2	31.75	28.33	23.94
3	26.51	23.51	20.00
4	28.06	25.18	21.36
5	27.23	24.34	20.61
6	14.87	13.62	11.30
7	54.74	47.76	40.89
8	25.44	22.27	19.40
9	44.53	39.53	34.94
10	35.72	32.65	28.07
11	47.10	40.74	36.04
12	38.95	34.72	30.21
Mean	34.39	30.56	26.29
Standard deviation	11.01	9.50	8.43

Note: PR denotes the reflector-equipped prototype, P0 denotes the prototype without a reflector, and C denotes the commercial heater. Each observation represents one simultaneous daily test conducted between 06:25 and 18:50 local time. Standard deviations are sample standard deviations.

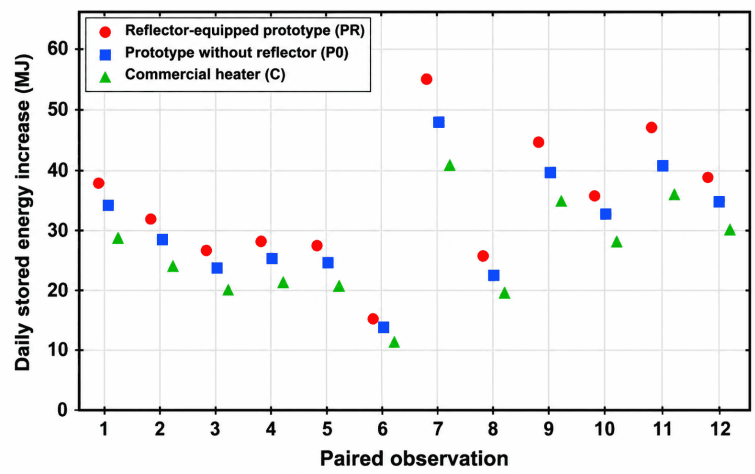


Figure 8. Daily increases in stored thermal energy for PR, P0, and C across the 12 matched observations

Note: PR denotes the reflector-equipped prototype, P0 denotes the prototype without a reflector, and C denotes the commercial heater.

The mean daily stored-energy increases were 34.39 MJ for PR, 30.56 MJ for P0 and 26.29 MJ for C. The corresponding sample standard deviations were 11.01, 9.50 and 8.43 MJ, respectively. These values describe the variation in stored-energy gains among the retained experimental days.

The daily energy gains ranged from 14.87 to 54.74 MJ for PR, from 13.62 to 47.76 MJ for P0, and from 11.30 to 40.89 MJ for C. All three heaters recorded their lowest energy gains in observation 6 and their highest in observation 7. This common pattern is consistent with the simultaneous exposure of the systems to changing outdoor conditions.

The cumulative stored-energy increases over the 12 separate test intervals were 412.67 MJ for PR, 366.70 MJ for P0 and 315.45 MJ for C. These totals represent the sums of the individual daily gains rather than continuous energy accumulation over the entire experimental campaign.

The results show that PR maintained a higher stored-energy gain across both lower- and higher-energy test days. The direction of the PR–P0 difference remained unchanged throughout the retained observations, providing a consistent experimental basis for evaluating the reflector assembly.

3.4 Stored-Energy Improvement Associated With the Reflector Assembly

The primary performance comparison was based on the paired daily differences between PR and P0. As shown in Figure 9, PR stored more thermal energy than the unmodified prototype in all 12 observations. The absolute daily difference ranged from 1.25 to 6.98 MJ, with a mean of 3.83 MJ and a sample standard deviation of 1.60 MJ. The sum of the additional stored-energy gains across the retained test intervals was 45.97 MJ.

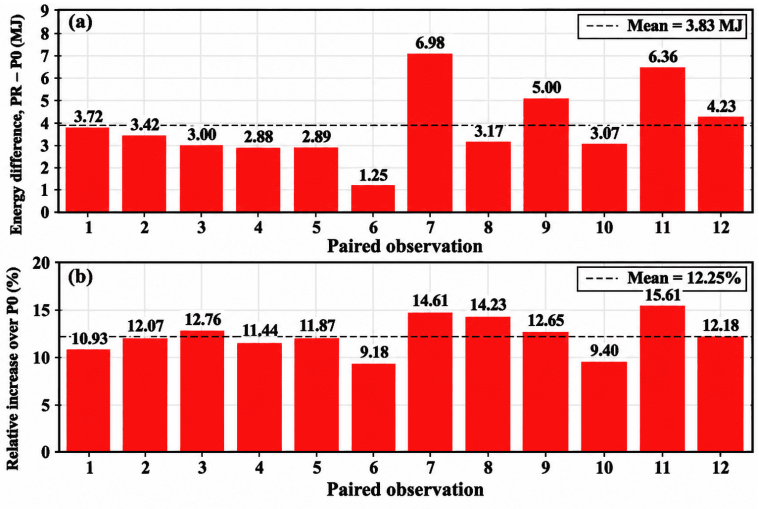


Figure 9. Paired comparison of daily stored thermal energy gains between PR and P0: (a) absolute energy differences and (b) relative increases over P0 across 12 matched observations

Note: PR denotes the reflector-equipped prototype and P0 denotes the prototype without a reflector. Dashed lines indicate the corresponding arithmetic means.

The overall relative increase, calculated from the mean daily stored-energy gains, was 12.54%. The individual daily increases ranged from 9.18% to 15.61%, with an arithmetic mean of 12.25% and a sample standard deviation of 1.94 percentage points. The distinction between the overall relative increase and the mean daily percentage arises from the different weighting of the observations. The overall indicator relates the total additional stored energy to the total energy stored by P0, whereas the arithmetic mean assigns equal weight to each daily percentage.

Observation 7 produced the largest absolute difference of 6.98 MJ, while observation 11 recorded the highest relative increase of 15.61%. Observation 6 had both the smallest absolute difference, at 1.25 MJ, and the smallest relative increase, at 9.18%. The results therefore show that the additional stored energy varied in magnitude across the experimental days, while PR maintained a positive advantage throughout the dataset.

The statistical comparison confirmed the consistency of the observed difference. The two-sided paired Student's t-test yielded $t(11) = 8.29$ ($p = 4.64 \times 10^{-6}$). The 95% confidence interval for the mean paired difference was 2.81–4.85 MJ.

An exact two-sided sign test gave $p = 0.000488$, with all 12 paired differences being positive. The sign-test result provides an additional assessment of the direction of the observed difference without requiring the paired differences to follow a normal distribution.

These statistical results describe the variation among the 12 retained experimental days. The paired t-test and its confidence interval are based on the assumption of approximately independent daily differences, with the t-based analysis additionally relying on an approximately normal distribution of those differences. The confidence interval represents between-day statistical variation under these assumptions and is distinct from uncertainty associated with sensor calibration, tank-temperature sampling or differences between manufactured units.

Taken together, the paired comparisons establish a consistent increase in daily stored thermal energy associated with the reflector-equipped configuration during the tested operating conditions. The magnitude of this increase is represented by the mean paired difference of 3.83 MJ and the overall relative gain of 12.54%.

3.5 Thermal Performance Interpretation and Comparison With Previous Research

The paired observations provide a basis for interpreting the thermal response of PR in relation to the collector geometry and storage conditions. The consistently greater stored-energy gains in PR indicate that the plate-and-foil assembly was associated with improved thermal energy accumulation over the monitored daily intervals. Since both prototypes shared the same nominal collector geometry and water inventory, the comparison focuses on the external reflector modification rather than differences in the principal collector and storage parameters.

The additional solar radiation redirected toward the evacuated tubes offers a physical explanation for the greater stored-energy gain. Reflector performance depends on the relative geometry of the reflective surface and the collector, including their dimensions, orientation and spacing. Previous investigations have examined reflector geometry and its influence on solar radiation interception and thermal performance in solar water heaters and related collector configurations [13–16]. These studies provide a basis for interpreting the reflector arrangement investigated here, although the collector structures and thermal performance indicators differ. The optical behavior of aluminum foil also depends on surface finish and the angular distribution of reflected radiation, as reported by Pozzobon et al. [18]. In the present configuration, the foil-covered steel plate was installed beneath the evacuated-tube array, with the brighter surface facing the tubes. The measured energy difference therefore represents the response of the complete reflector assembly rather than a separately measured contribution from the foil's optical properties.

The temperature profiles provide further insight into the thermal response of the three systems. Both experimental prototypes exhibited smaller temperature differences between the monitored tank heights than C, while PR reached higher late-day temperatures than P0. The observed temperature distributions are consistent with the interaction between thermosiphon circulation and thermal stratification in water-in-glass evacuated-tube systems [3, 4]. The more closely grouped temperatures in the experimental prototypes indicate greater thermal uniformity at the sampled locations. At the same time, the different storage capacities of the prototypes and C explain why their temperature rankings did not necessarily correspond to their stored-energy rankings.

The distinction between stored thermal energy and collector efficiency is important when comparing the present findings with previous research. Studies of evacuated-tube collectors incorporating phase change materials, modified heat transfer arrangements and integrated storage components have reported improvements using different thermal performance indicators [9–12]. The present investigation instead evaluates the increase in sensible thermal energy retained in the water tank during a fixed daily interval. Accordingly, the 12.54% increase represents the relative difference in daily stored-energy gains between PR and P0. It should not be interpreted as an equivalent increase in collector efficiency or directly compared with improvements in phase change storage performance reported for other collector configurations.

C provides an additional perspective on system-level performance. Its mean daily stored-energy gain was 26.29 MJ, compared with 30.56 MJ for P0 and 34.39 MJ for PR. PR therefore exceeded C by an average of 8.10 MJ per test day, equivalent to an overall relative difference of 30.82%. However, the commercial system had a smaller water inventory of 263 L and differed from the experimental prototypes in tube spacing and inclination. These differences contributed to the overall performance comparison, making the PR–P0 contrast the appropriate basis for assessing the reflector assembly.

The findings also distinguish the present experimental approach from studies involving multiple collector modifications. Previous investigations have examined changes to internal heat transfer structures, collector geometry and thermal storage arrangements as means of improving solar thermal performance [7–10]. In the present study, the nominal collector geometry, water inventory and natural-circulation arrangement were maintained between the two experimental prototypes, while the external reflector assembly constituted the principal intentional modification. This design provided a focused comparison of daily stored-energy gains without combining the reflector with changes to the internal collector structure.

Overall, the results establish a consistent positive association between the installed reflector assembly and the daily sensible thermal energy accumulated in the storage water. The paired comparison demonstrates the relevance of evaluating reflector-assisted solar water heaters through stored-energy gains rather than collector temperature alone. By connecting an external collector modification with a directly measured storage-related performance indicator, the study provides experimental evidence for assessing practical approaches to improving solar heat utilization in water-in-glass evacuated-tube systems.

3.6 Implications for Sustainable Solar Thermal Energy Utilization

The experimental results demonstrate the relevance of reflector-assisted configurations to the improvement of thermal energy storage in water-in-glass evacuated-tube solar water heaters. Across the 12 paired observations, PR accumulated an average of 3.83 MJ more thermal energy per test day than P0. This difference corresponded to approximately 1.06 kWh of additional stored thermal energy during the daily monitoring interval. The overall relative increase of 12.54% indicates that the reflector assembly contributed to greater thermal energy accumulation without changes to the nominal tube arrangement, collector inclination or water inventory.

From an engineering perspective, the principal advantage of the investigated configuration lies in its relatively simple construction. The reflector consisted of an aluminum-foil-covered steel plate installed beneath the existing evacuated-tube array. Recent studies have examined solar water heating improvements through changes to collector construction, internal heat transfer arrangements and thermal storage components [7–10]. In contrast, the modification investigated here was confined to the external reflector assembly, while the nominal collector geometry, natural-circulation arrangement and water inventory remained unchanged. The experimental comparison therefore illustrates

how adjustments to the external collector configuration can influence the thermal energy retained in a solar water heating system.

The results are particularly relevant to solar thermal applications in which the amount of heat stored during daylight hours determines the availability of hot water later in the day. Increasing the daily stored-energy gain can contribute to more effective use of the solar resource and support the role of solar water heating in sustainable thermal energy supply. The measured improvement provides a quantitative basis for considering reflector assemblies in the design and modification of evacuated-tube solar water heaters.

The performance difference must nevertheless be interpreted in relation to the adopted experimental conditions. The additional 1.06 kWh represents thermal energy accumulated in the storage water over the monitored interval, rather than an equivalent reduction in electricity consumption. Actual displacement of conventional energy depends on hot-water demand, delivery temperature, auxiliary heating arrangements and subsequent storage losses. Similarly, the reported improvement describes the tested reflector assembly and should not be interpreted as a direct measurement of collector efficiency.

The engineering relevance of the reflector assembly is therefore reflected in the additional sensible thermal energy retained during the monitored heating period. This storage-based assessment provides a practical measure for comparing external collector modifications under equivalent operating conditions and supports the evaluation of solar water heating systems intended for renewable heat supply.

3.7 Experimental Scope and Interpretation of the Results

The paired experimental arrangement provides a consistent basis for comparing the thermal response of the two prototypes. Both systems were operated simultaneously, with the same nominal collector geometry and a water inventory of 316 L. Previous modeling and experimental investigations have identified collector geometry, inclination and circulation characteristics as important factors in the thermal response of water-in-glass evacuated-tube systems [4], [19]. Maintaining the same nominal values of these parameters in the two experimental prototypes therefore provided a more focused comparison of the installed reflector assembly. The positive PR–P0 difference in all 12 observations further established the consistency of the measured storage improvement across the retained experimental days.

The experimental design involved one physical unit of each configuration. Accordingly, the results characterize the performance of the installed reflector assembly on the tested prototype rather than variation among independently manufactured units. The addition of the reflector was the only intentional construction difference, although the experiment did not include a crossover arrangement in which the reflector was transferred between prototypes. The reported difference is therefore associated with the complete reflector-equipped configuration.

The stored-energy assessment was based on the arithmetic mean of temperatures measured at three vertical positions in each tank. This procedure was applied consistently across the three heaters and provided a common basis for estimating their daily energy gains. Because temperature stratification affects the distribution of thermal energy within a storage tank, the three-point mean represents an approximation of the volume-averaged water temperature. The corresponding stored-energy values should be interpreted within this measurement framework.

The statistical results provide additional information about the consistency of the paired differences. The mean difference of 3.83 MJ had a 95% Student's t confidence interval of 2.81–4.85 MJ. The paired t-test yielded $t(11)$ = 8.29 and $p = 4.64 \times 10^{-6}$, while the exact two-sided sign test gave p = 0.000488. These results describe the variation and direction of the differences across the 12 retained observations, conditional on independence between experimental days. They do not incorporate the full measurement uncertainty of the temperature acquisition and stored-energy calculations.

The reflector's solar-weighted reflectance, angular reflection characteristics and thermal emittance were not measured separately. The observed performance difference therefore represents the combined effect of the installed plate-and-foil assembly, including its optical and geometrical interaction with the evacuated-tube array. The available numerical observations establish the daily stored-energy difference, while the temperature profiles provide complementary information about the thermal response of the systems.

Taken together, the results provide a quantitative experimental assessment of a reflector-assisted water-in-glass evacuated-tube solar water heater under outdoor operating conditions. The combination of matched daily observations, temperature profiles and paired statistical analysis establishes the magnitude and consistency of the storage improvement within the investigated configuration. The findings also provide a reference for the design of subsequent experiments involving alternative reflector geometries, independently manufactured prototypes and representative hot-water withdrawal conditions.

4 Conclusions

This study investigated the effect of an aluminum-foil-covered steel reflector on daily thermal energy storage in water-in-glass evacuated-tube solar water heaters. Three systems were tested simultaneously in Trujillo, Peru,

including two experimental prototypes with identical nominal collector geometry and a water inventory of 316 L each, together with a commercial reference heater. PR incorporated a 6.624 m^2 aluminum-foil-covered steel plate beneath the evacuated-tube array. The performance assessment was based on 12 matched daily observations obtained over a common monitoring interval of 06:25–18:50.

The principal findings are summarized as follows:

(1) PR recorded greater daily stored-energy gains than P0 in all 12 matched observations. The mean daily gains were 34.39 and 30.56 MJ, respectively, giving a mean paired difference of 3.83 MJ and an overall relative increase of 12.54%. Individual daily increases ranged from 9.18% to 15.61%.

(2) The paired statistical analysis confirmed a consistent positive difference across the retained observations. The 95% confidence interval for the mean paired difference was 2.81–4.85 MJ. The paired t-test yielded $t(11) = 8.29$ and $p = 4.64 \times 10^{-6}$, while the exact two-sided sign test produced $p = 0.000488$. These results characterize the between-day variation in stored-energy gains under the adopted experimental design.

(3) The daily temperature profiles showed higher late-day temperatures in PR than in P0. Both experimental prototypes also exhibited smaller temperature differences between the monitored tank heights than C. The combined temperature and energy analyses demonstrated the importance of accounting for water inventory and temperature change when comparing the thermal performance of solar water heating systems.

(4) C recorded a mean daily stored-energy gain of 26.29 MJ, compared with 34.39 MJ for PR. The corresponding overall difference of 30.82% reflected the combined effects of different storage capacities, tube spacing, collector inclinations and reflector configurations. The comparison between the two nominally identical experimental prototypes therefore provided the more direct assessment of the reflector-associated storage improvement.

(5) The mean additional stored thermal energy of 3.83 MJ corresponded to approximately 1.06 kWh per test day. This result demonstrates the practical potential of an aluminum-foil-covered reflector assembly to increase thermal energy accumulation in evacuated-tube solar water heaters without modifying their basic natural-circulation arrangement. The improvement is relevant to the development of solar thermal technologies that make more effective use of renewable heat for domestic water heating.

Overall, the experimental findings establish a consistent increase in daily stored thermal energy associated with the installed reflector assembly under the investigated conditions. The study contributes a paired experimental assessment that distinguishes stored-energy gains from collector efficiency and separates the reflector comparison from broader differences between complete heater configurations.

Author Contributions

Conceptualization, Y.R.L.C. and E.M.M.O.; methodology, Y.R.L.C.; software, Y.R.L.C.; validation, Y.R.L.C., E.M.M.O., and N.R.C.; formal analysis, Y.R.L.C. and M.A.K.; investigation, Y.R.L.C. and E.M.M.O.; resources, E.M.M.O. and N.R.C.; data curation, Y.R.L.C.; writing—original draft preparation, Y.R.L.C.; writing—review and editing, Y.R.L.C., E.M.M.O., N.R.C., and M.A.K.; visualization, Y.R.L.C.; supervision, E.M.M.O. and N.R.C.; project administration, E.M.M.O.; funding acquisition, N.R.C. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] T. Sood, D. Lee, R. Kumar, and Ü. Ağbulut, "A detailed review of evacuated tube solar collectors for sustainable energy, current trends and future prospects," *Sol. Energy*, vol. 303, p. 114054, 2026. https://doi.org/10.1016/j.solener.2025.114054

[2] Y. Shen, Y. Yang, B. Liu, Y. Liu, Q. Zhao, Y. Ma, M. Zhou, T. Wang, L. Guo, and W. Que, "Evacuated collector tube-based solar energy conversion system: Device structure, practical application, and sustainability assessment," *Appl. Energy*, vol. 402, p. 126921, 2025. https://doi.org/10.1016/j.apenergy.2025.126921

[3] G. L. Morrison, I. Budihardjo, and M. Behnia, "Measurement and simulation of flow rate in a water-in-glass evacuated tube solar water heater," *Sol. Energy*, vol. 78, no. 2, pp. 257–267, 2005. https://doi.org/10.1016/j.solener.2004.09.005

[4] E. M. Mendoza Orbegoso, J. Alva Alcántara, L. Julca Verástegui, J. C. Bengoa, D. Marcelo-Aldana, R. La Madrid Olivares, and K. G. Kyprianidis, "Thermofluidics in water-in-glass evacuated-tube solar collectors analysis based on the symmetry conditions of heat flux and tilt angle," *Symmetry*, vol. 17, no. 1, p. 44, 2025. https://doi.org/10.3390/sym17010044

[5] J. Gong, Z. Sun, J. Wang, and P. D. Lund, "Performance studies of novel all-glass heat pipe evacuated collector tube integrating numerical simulation and experiment method," *Sol. Energy*, vol. 253, pp. 491–500, 2023. https://doi.org/10.1016/j.solener.2023.02.028

[6] M. Yang, L. Zhi, K. Diao, Y. Zhu, and R. A. Taylor, "Optical and thermal performance analysis of a compact solar collector with heat-pipe evacuated tube," *Sol. Energy*, vol. 258, pp. 118–129, 2023. https://doi.org/10.1016/j.solener.2023.04.036

[7] D. Zhang, Y. Diao, Z. Wang, Y. Pan, M. Sun, X. Wang, P. Du, and Y. Zhao, "Thermal performance of two evacuated tube solar collectors with flat heat pipes," *Appl. Therm. Eng.*, vol. 241, p. 122366, 2024. https://doi.org/10.1016/j.applthermaleng.2024.122366

[8] A. A. Khadom, H. B. Mahood, A. A. Mahmmod, Q. Hassan, and H. A. Kazem, "Improving solar water heating performance and reducing emissions by evacuated tube collectors with preheating units: Iraq as a case study," *Appl. Therm. Eng.*, vol. 265, p. 125596, 2025. https://doi.org/10.1016/j.applthermaleng.2025.125596

[9] A. Uniyal, Y. K. Prajapati, and D. Kumar, "Experimental investigation of wire-mesh equipped U-tube evacuated tube collector integrated with phase change material for hot water generation," *Appl. Energy*, vol. 377, p. 124501, 2025. https://doi.org/10.1016/j.apenergy.2024.124501

[10] A. Elbrashy, A. Elshennawy, M. Rashad, and M. El-fakharany, "Assessment of thermal behavior for storage material filled in annulus coaxial aluminum pipes and inserted in evacuated tube collector: An experimental and mathematical investigation," *J. Energy Storage*, vol. 105, p. 114674, 2025. https://doi.org/10.1016/j.est.2024.114674

[11] C. Zhu, X. Dong, J. Wang, Q. Luo, S. Yan, C. Li, Z. Zhu, Q. Liu, and Y. Huang, "Experimental thermal performance investigation of air-type solar collector with PCM-rod embedded in vacuum tube," *J. Energy Storage*, vol. 91, p. 112060, 2024. https://doi.org/10.1016/j.est.2024.112060

[12] S. Mellouli, F. Askri, T. Alqahtani, and S. Algarni, "Energy storage enhancement in an innovative photovoltaic-thermal collector utilizing evacuated tubes and phase change materials," *J. Energy Storage*, vol. 154, p. 121360, 2026. https://doi.org/10.1016/j.est.2026.121360

[13] S. Mandal and S. K. Ghosh, "Experimental investigation of the performance of a double pass solar water heater with reflector," *Renew. Energy*, vol. 149, pp. 631–640, 2020. https://doi.org/10.1016/j.renene.2019.11.160

[14] N. N. Nikolić, N. S. Lukić, A. Nešović, and D. M. Nikolić, "Optimal size and position of the planar back reflector moveable only in the direction normal to the bifacial solar collector plane," *Therm. Sci.*, vol. 28, no. 6, pp. 4483–4497, 2024. https://doi.org/10.2298/TSCI240123101N

[15] M. B. Farghaly, R. N. Alahmadi, H. H. Sarhan, and E. S. Abdelghany, "Experimental study of simultaneous effect of evacuated tube collectors coupled with parabolic reflectors on traditional single slope solar still efficiency," *Case Stud. Therm. Eng.*, vol. 49, p. 103304, 2023. https://doi.org/10.1016/j.csite.2023.103304

[16] M. Sheikholeslami and W. M. M. Al-Jammasi, "Integrated U-tube solar collector with nanoparticle-enhanced paraffin and parabolic reflector for improved thermal storage performance," *J. Energy Storage*, vol. 156, p. 121447, 2026. https://doi.org/10.1016/j.est.2026.121447

[17] J. Zhu, C. Zhang, F. Chen, Y. Zhao, X. Zhang, and H. Li, "Enhancing thermal energy collection performance for non-imaging concentrating solar system with evacuated tube absorber by heat storage rods," *Appl. Therm. Eng.*, vol. 268, p. 125908, 2025. https://doi.org/10.1016/j.applthermaleng.2025.125908

[18] V. Pozzobon, W. Levasseur, K.-V. Do, B. Palpant, and P. Perré, "Household aluminum foil matte and bright side reflectivity measurements: Application to a photobioreactor light concentrator design," *Biotechnol. Rep.*, vol. 25, p. e00399, 2020. https://doi.org/10.1016/j.btre.2019.e00399

[19] E. M. Mendoza Orbegoso, J. Alva Alcántara, L. Julca Verástegui, E. Asto-Rodriguez, J. C. Bengoa Seminario, Y. Leiva Calvanapón, J. Silvera Otañe, D. Marcelo-Aldana, and R. La Madrid Olivares, "Predictive modeling and validation of water in glass evacuated tube solar heaters," *Sol. Energy*, vol. 303, p. 114125, 2026. https://doi.org/10.1016/j.solener.2025.114125

[20] J. Alva Alcántara, E. Mendoza Orbegoso, N. R. Caetano, L. Julca Verástegui, J. Bengoa Seminario, J. Silvera Otañe, Y. Leiva Calvanapón, and G. Lorenzini, "A real-time IoT and cloud monitoring framework for performance enhancement of solar evacuated tube heaters," *Front. Heat Mass Transfer*, vol. 24, no. 1, p. 13, 2026. https://doi.org/10.32604/fhmt.2025.074995

[21] A. K. Singh, "Analysis for modified augmentation of solar desalination assisted water heating system: Energy-matrices and environ-economics," *Sol. Energy*, vol. 285, p. 113120, 2025. https://doi.org/10.1016/j.solener.2024.113120

[22] A. Agrawal and A. Kumar, "Experimental comparison and 6E analyses of double-ended evacuated tube collector based atmospheric water harvesting with and without PCM," *Sol. Energy Mater. Sol. Cells*, vol. 282, p. 113343, 2025. https://doi.org/10.1016/j.solmat.2024.113343

[23] A. Hemmatian, H. Kargarsharifabad, A. A. Esfahlani, N. Rahbar, and S. Shoeibi, "Improving solar still performance with heat pipe/pulsating heat pipe evacuated tube solar collectors and PCM: An experimental and environmental analysis," *Sol. Energy*, vol. 269, p. 112371, 2024. https://doi.org/10.1016/j.solener.2024.112371

[24] W. M. Hashim, I. S. Ahmed, A. K. Khlief, R. A. Jessam, and A. A. Jaddoa, "Performance evaluation of evacuated tube receiver at various flow rates under Baghdad climate with nanofluid as working fluids," *Energy Eng.*, vol. 122, no. 6, pp. 2485–2501, 2025. https://doi.org/10.32604/ee.2025.061630

[25] S. Farhadi, H. A. Seresht, H. Aghakhani, and B. Baghapour, "Performance prediction of heat pipe evacuated tube solar collectors: Analytical modeling and data-driven machine learning/ANN approach with developing web application," *Energy*, vol. 321, p. 135512, 2025. https://doi.org/10.1016/j.energy.2025.135512

[26] A. Ozsoy and M. Galip, "Thermal performance improvement of evacuated U-pipe solar collectors using passive elements," *Sol. Energy*, vol. 301, p. 113940, 2025. https://doi.org/10.1016/j.solener.2025.113940